

PROPERTIES OF FLUIDS

INTRODUCTION

- A matter exists in either the solid state or the fluid state. The fluid state is further divided into (i) liquid state, (ii) gaseous state. In fact, the matter exists in any of three states, i.e., solid, liquid, and gaseous state. For example, water, which ordinarily occurs in a liquid state, may also occur under natural conditions in a solid state as ice and in a gaseous state as vapour.
- In solids the molecules are very closely spaced, but in liquids the spacing between molecules is relatively large and in gases the space between the molecules is still larger. As such in a given volume, a solid contains a large number of molecules, a liquid contains relatively less number of molecules and a gas contains much less number of molecules. It is thus in solids the forces of attraction between the molecules is large. In liquids the force of attraction between molecules is relatively less and in gaseous, the force of attraction is much less.
- For mechanical analysis, a fluid is considered to be continuous, i.e., a continuous distribution of matter with no voids or empty spaces.
- Another difference between the solids and the fluids: their ability to resist the external forces.
 1. A solid can resist tensile, compressive and shear forces up to a certain limit.
 2. A fluid has no tensile strength or very little of it, and it can resist the compressive forces only when it is kept in containers. When subjected to a shear force, a fluid deforms continuously as long as that force is applied.

The two classes of fluids: gases and liquids also exhibit quite different characteristics.

- Gases can be compressed much readily under the action of external pressure and when external pressure is removed, gases tend to expand indefinitely, whereas, under ordinary conditions, liquids are only slightly compressible and therefore they may, for most purposes, be regarded as incompressible.

Definition of a FLUID

- A fluid may be defined as a substance which is capable of flowing. It has no definite shape of its own but conforms to the shape of the containing vessel. Further, even a small amount of shear force causes

the fluid to undergo continuous deformation which continues as long as the shear continues to be applied.

- A liquid is a fluid, which possesses a definite volume, which varies only slightly with temperature and pressure, since under ordinary conditions liquids are difficult to compress. So they may be regarded as incompressible.
- A gas is a fluid, which is compressible and possesses indefinite volume. The volume of gas varies with the variation in the state of conditions, since the change in temperature and pressure have a significant effect on its volume and pressure.

A vapour is a gas whose temperature and pressure are such that it is very near to the liquid state. Thus steam may be considered as a vapour because its state is normally not far from that of water.

Fluids are also classified as:

1. Ideal fluids
 2. Practical or real fluids
- **Ideal fluids** are those fluids which have no viscosity and surface tension, and are incompressible. However, in nature, such ideal fluids do not exist and therefore these are only imaginary fluids.
 - **Practical or real fluids** are those fluids which are actually available in nature. These fluids possess the properties such as viscosity, surface tension and compressibility, and therefore a certain amount of resistance is always offered by these fluids.

Units of Measurement

- Units may be defined as those standards in terms of which various physical quantities like length, mass, time, force, area, volume, velocity, acceleration, etc., are measured.
- The system of units is based on Newton's Second Law of Motion, which states that if a body of mass m moves under acceleration a , $F = m \times a$, where F is the force, m is mass and a is acceleration

There are in general four system of units

- 1). Two in metric (C.G.S or M.K.S) system
- 2). Two in (F.P.S) system.

of the two, one is known as an absolute or physical system.

- Others are gravitational or engineering systems.

The difference between the absolute and gravitational systems is that in the former the standard is the unit of mass. The unit of force is then derived by Newton's law.

In the gravitational system the standard is the unit of force, and the unit of mass is derived by Newton's law.

Quantity	Metric Units		English Units	
	Gravitational	Absolute	Gravitational	Absolute
Length	Meter (m)	Meter (m)	Foot (ft)	Foot (ft)
Time	Second (Sec)	Second (Sec)	Second (Sec)	Second (Sec)
Mass	Metric Slug (msl)	Gram [gm (mass)]	Slug (sl)	Pound [lb (mass)]
Force	Kilogram [Kg (F)]	Dyne	Pound [lb (F)]	Poundal (Pal)
Temperature	$^{\circ}\text{C}$	$^{\circ}\text{C}$	$^{\circ}\text{F}$	$^{\circ}\text{F}$

The table below illustrates all the four systems of units in which the units are defined, so that one unit of force equals one unit of mass times one unit of acceleration.

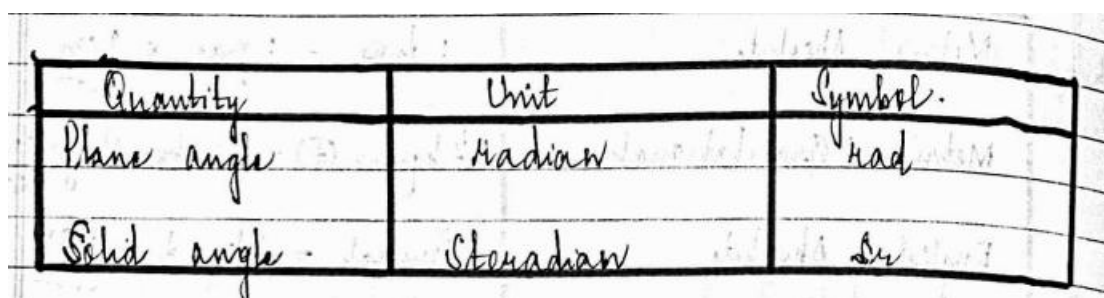
Systems of units	Relationships
Metric Absolute	$1 \text{ dyne} = 1 \text{ gram} \times \frac{1 \text{ cm}}{\text{Sec}^2}$
Metric Gravitational	$1 \text{ kilogram (F)} = 1 \text{ metric slug} \times \frac{1 \text{ m}}{\text{Sec}^2}$
English Absolute	$1 \text{ Poundal} = 1 \text{ Pound} \times \frac{1 \text{ ft}}{\text{Sec}^2}$
English Gravitational	$1 \text{ Pound} = 1 \text{ Slug} \times \frac{1 \text{ ft}}{\text{Sec}^2}$

The use of different systems of units by the scientists and the engineers and also by the different countries of the world often leads to a lot of confusion. Therefore, a common system of units known as International System of Units, abbreviated as SI units. More and more countries of the world are now adopting this system of units.

There are six base units in SI system

quantity	unit	symbol
Length m	meter	m
Mass	kilogram	Kg
Time	Second	S
Electro current	Ampere	A
Thermodynamic temperature	Kelvin	K
luminous intensity	candela	Cd

In addition to the above noted leads units are two supplementary units



Quantity	Unit	Symbol
Plane angle	radian	rad
Solid angle	steradian	sr

In SI units the unit of force is newton (N), which according to Newton's second law of motion is expressed $1 \text{ N} = 1 \text{ kg} \times 1 \text{ m/s}^2$.

i.e., a force of 1 N required to accelerate a mass of 1 kg by 1 m/s^2 .

In using SI units certain rules and conventions are to be followed:

- I. Names of units, even when they are named after persons, are not written with the first letter capital when written in the spelled form.
e.g., newton, joule, watt etc.
- II. The symbols for the units which are named after persons are written with capital first letter of the name.
e.g., N for newton, J for joule, W for watt etc.
- III. The symbols for all other units are written in lower case (small letters).
e.g., m for metre, s for second, kg for kilogram etc.
- IV. A dot is inserted in the spaces between the symbols for the compound units.
e.g., N·m for newton metre, kW·h for kilowatt hour etc.
- V. Numbers with more than three digits should be written in groups of three with narrow spaces in between, consecutive group however A sequence of four digits is, usually not broken.

Examples:

- 50000 should be written as 50 000

- 72345.685 should be written as 72 345.685
- 0.13579 should be written as 0.135 79
- 9810 may should be written as 9 810 or 9810

(vi) The decimal multiples and sub-multiples of the units are formed by using the prefixes.

1. Mass Density (Specific Mass) (ρ)

Of a fluid is the mass which it posses per unit volume. It is denoted by symbol ρ (rho).

- In SI units it is expressed in kg/m^3 .
- In metric gravitational systems it is metric slug/ m^3
- Metric absolute systems it is metric gm/cm^3
- English Gravitational system it is : sluge/ ft^3
- English absolute systems it is pound/ ft^3 , lb. ft^3 .

The mass density of water at 4°C in different units is 1000 kg/m^3 (or) 102 mer/m^3 , (or)

1 gm/cc (or) 1.94 slugs/m^3 (or) 62.4 lb/ft^3

2. Specific Weight (Weight Density) (w or γ (gama))

of a fluid is its weight it possesses per unit volume. It is denoted by symbol w or γ

In SI units: N/m^3

- Metric Gravitational units = kilogramf/ m^3
- Metric absolute units = dynos/ cm^3
- English Gravitational units = lb/ft^3
- English Absolute units = pdl /ft^3

The specific weight of water at 4°C in different units:

- 9810 N/m^3 (9.81 kN/m^3) or
- 1000 kgf/m^3 or
- 981 dynos /cm^3 or
- 62.4 lb(F)/ft^3 or

- $(62.4 \times 32.2) \text{ plb/ft}^3$

The mass density (ρ) and specific weight (w) are related as:

$$w = \rho g \text{ or } \rho = w/g$$

Where g = acceleration due to gravity.

3. specific volume:

Of a fluid is the volume of the fluid unit weight. Thus it is reciprocal of specific it is denoted by γ .

In S.I. units..... m^3/N

Metric absolute units..... m^3/kg (f)

English gravitational unit..... $\text{ft}^3/\text{lb}(f)$

English absolute unit..... ft^3/pdl

4. Specific Gravity

Specific gravity (S_g or G) is the ratio of specific weight (or mass density) of a fluid to the specific weight (or mass density) of a standard fluid.

- The standard fluid chosen for comparison is pure water at 4°C (39.2°F).
- For gases, the standard fluid chosen is either hydrogen or air.
- The specific gravity of water at 4°C is 1 .
- The specific gravity of mercury = 13.5 to 13.6 .

Knowing the specific gravity of any liquid, its specific weight can be calculated as:

$$w = \text{S.g. of liquid} \times \text{Specific weight of water} \\ = (\text{S.g. of liquid}) \times 9810 \text{ N/m}^3$$

Equation of State: The Perfect Gas

The density ' ρ ' of a particular gas is related to its absolute pressure ' p ' and absolute temperature ' T ' by the equation of state, which for a perfect gas takes the form:

$$p = \rho RT \text{ or } pV = mRT \dots (1)$$

In which ' R ' is a constant called the gas constant, the value of which is constant for the gas concerned, and V is the volume occupied by the mass ' m ' of the gas.

The absolute pressure is the pressure measured above absolute zero (or complete vacuum) and is given by:

$$P_{\text{abs}} = P_{\text{gage}} + P_{\text{atm}}$$

The absolute temperature is expressed in kelvin, i.e., K , when the temperature is measured in $^{\circ}\text{C}$.

No actual gas is perfect. However, most gases obey this relation closely and hence their pressure, density, and temperature may, to a good approximation, be related by equation (1).

Similarly air at normal temperature and pressure behaves closely with the equation of state.

It may be noted that the gas constant ' R ' is defined by eq (1) as $p/\rho T$ and therefore, its dimensional expressions is $(FL/M\theta)$

Thus in S.I. units the gas constant ' R ' is expressed in newton-metre per kilogram per kelvin, i.e. $(N \cdot W/\text{kg} \cdot K)$. But 1 joule = 1 newton $\times$ 1 metre

$$R = \text{joule}/(\text{kilogram}/\text{kelvin})$$

Again, $1 \text{ N} = 1 \text{ kg} \times 1 \text{ m}/\text{s}^2$

$$R = \frac{m^2}{s^2 \cdot K}$$

Since specific volume may be defined as reciprocal of mass density, the equation of state may also be expressed in terms of specific volume of the gas:

$$pv = RT$$

In which v is specific volume.

The equation of state may also be expressed as:

$$p = \gamma RT$$

in which γ is the specific weight of gas.

The unit for the gas constant (R) then becomes ($\text{m}^2/\text{s}^2 \cdot \text{K}$).

For a given temperature and pressure, equation(1) indicate that $pR = \text{constant}$.

By Avogadro's hypothesis, all pure gases at the same temperature and pressure have the same number of molecules per unit volume. The density is proportional to the mass of an individual molecule, and so the product of R and the molecular weight ' M ' is constant for all perfect gases. This product MR , know as universal gas constant.

Viscosity

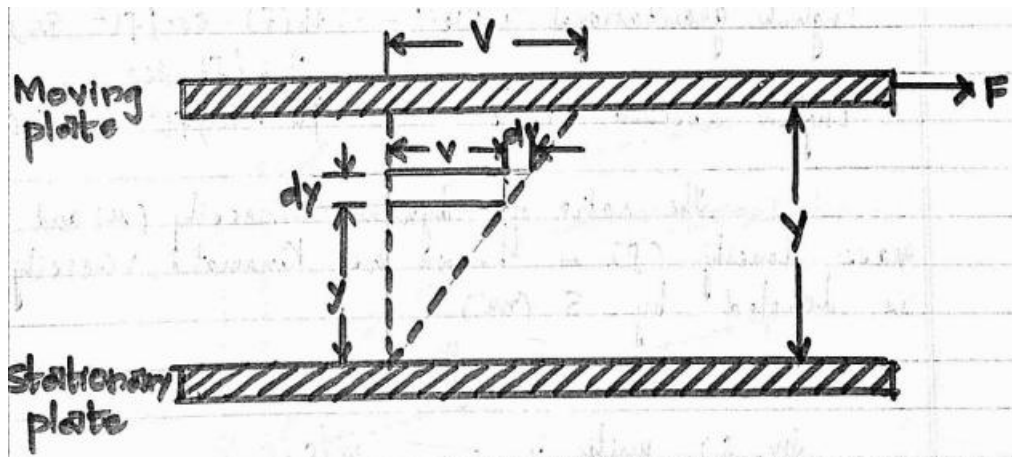
Viscosity is that property of a fluid by virtue of which it offers resistance to the movement of one layer of fluid over an adjacent layer. It is primarily due to cohesion and molecular momentum exchange between fluid layers, and as flow occurs, these effects appear as shearing stresses between the moving layers of fluid.

Consider two plates sufficiently large placed a small distance Y apart, the force between them filled with fluid. The lower plate is assumed to be at rest while the upper one is moved parallel to it with a velocity V by the application of a force F corresponding to area A of the moving plate in contact with the fluid.

Particles of the fluid in contact with each plate will adhere to it, and if the distance Y and velocity V are not too great, the velocity ' γ ' at a distance ' y ' from the lower plate will vary uniformly from zero at the lower plate (which is at rest) to V at the upper moving plate.

Experiments show that for a large variety of fluids:

$$F \sim \frac{AV}{Y}$$



It may be seen from similar triangles in fig. that the ratio V/Y can be replaced by the velocity gradient (dv/dy) , which is the rate of angular deformation of the fluid.

If a constant of proportionality μ (Greek "mu") be introduced, the shear stress τ ("tau") equal to F/A between any two thin sheets of fluid may be expressed as:

$$\tau = \frac{F}{A} = \mu \frac{V}{Y} = \mu \frac{dv}{dy} \dots (2)$$

Equation (2) is called Newton's Equation of Viscosity:

$$\mu = \frac{\tau}{dv/dy}$$

where μ is called the coefficient of viscosity (or) dynamic viscosity) or simply viscosity.

Thus dynamic viscosity (μ) may be defined as the shear stress required to produce unit rate of angular deformation.

In SI units, μ is expressed in $\text{N} \cdot \text{s}/\text{m}^2$ or $\text{kg}/\text{m} \cdot \text{s}$.

- Metric gravitational units..... $\text{kg}(F) - \text{sec}/\text{m}^2$ or $\text{NSL}/\text{w} \cdot \text{sec}$
- Metric absolute units..... $\text{dyne} - \text{sec}/\text{cm}^2$ or $\text{Gm}(\text{mass})/\text{cm} - \text{sec}$ which is also called "poise"; one hundredth of poise is "centipoise".
- English gravitational units.... $\text{lb}(F) \cdot \text{sec}/\text{ft}^2$ or $\text{slug}/\text{ft} \cdot \text{sec}$
- English absolute units.... $\text{pdl} - \text{sec}/\text{ft}^2$ (or) $\text{lb}(m)/\text{ft} - \text{sec}$

The ratio of dynamic viscosity (μ) and the mass density (ρ) is known as Kinematic viscosity and is denoted by $\gamma(m')$

$$\gamma = \frac{\mu}{\rho}$$

In S.I units $\gamma \dots m^2/s$

metric unit cm^2/sec or m^2/sec .

The unit cm^2/sec is termed as stoke;

one-hundredth part is centistoke.

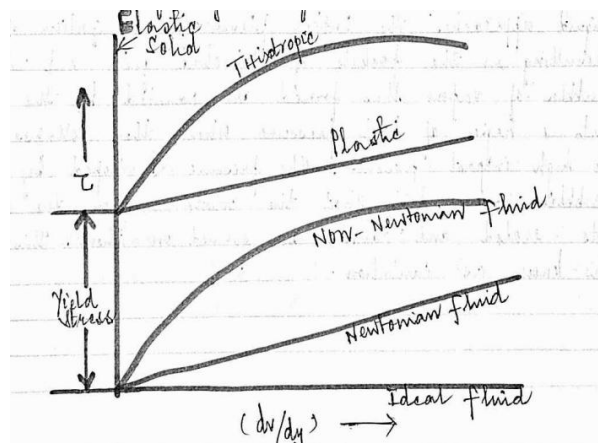
In English system $\rightarrow ft^2/sec$.

Common fluids such as air, water, glycerine, kerosene etc. follow eq. (2).

There are certain fluids which, however, do not follow Newton's law of viscosity. Accordingly, fluids may be classified as Newtonian fluids and Non-Newtonian fluids.

- In a Newtonian fluid there is a linear relation between the magnitude of shear stress and the resulting rate of deformation. i.e., The constant of proportionality μ in equation (2) does not change with rate of deformation.
- In a Non-Newtonian fluid there is a non-linear relation between the magnitude of the applied shear stress and the rate of angular deformation. In the case of a plastic substance which is a non-Newtonian fluid, an initial yield stress is to be exceeded to cause a continuous deformation.

An ideal plastic has a definite yield stress and a constant linear relation between shear stress and rate of angular deformation.



A thixotropic substance, which is a non-Newtonian fluid, has a non-linear relationship between the shear stress and rate of angular deformation, beyond an initial yield stress. The printer's ink is an example for thixotropic fluid.

An ideal fluid is defined as that having zero viscosity or, in other words, shear stress is always zero regardless of the motion of the fluid. Thus an ideal fluid is represented by the horizontal axis ($\tau = 0$).

Vapour Pressure

All liquids possess a tendency to evaporate or vaporize, i.e., to change from the liquid to the gaseous state. Such vaporization occurs because of continuous escaping of the molecules through the free surface. When the liquid is confined in a closed vessel, the ejected vapour molecules get accumulated in the space between the free liquid surface and the top of the vessel. This accumulated vapour of the liquid exerts a partial pressure on the liquid surface, which is known as vapour pressure of the liquid.

If in any flow system the pressure at any point in the liquid approaches the vapour pressure, vaporization of liquid starts, resulting in the pockets of dissolved gases and vapours. The bubbles of vapour thus formed are carried by the flowing liquid into a region of high pressure where they collapse, giving rise to high impact pressure. The pressure developed by the collapsing bubbles is so high that the material from the adjoining boundary gets eroded and cavities are formed on them. This phenomenon is known as cavitation.

Compressibility & Elasticity

All fluids may be compressed by the application of external force and when the external force is removed, the compressed volume of fluids expands to their original volume. Thus fluids also possess elastic characteristics like elastic solids.

Compressibility (K) of a fluid is quantitatively expressed as inverse of the bulk modulus of elasticity K , of the fluid, which is defined as:

$$K = \frac{\text{Stress}}{\text{Strain}} = -\frac{dp}{(dV/V)} = \frac{\text{Change in pressure}}{\left(\frac{\text{Change in volume}}{\text{Original volume}}\right)}$$

The bulk modulus of elasticity K is a measure of the incremental change in pressure dp which takes place when a volume V of the fluid is changed by an incremental amount dV .

Since a rise in pressure always causes a decrease in volume, dV is always negative, and minus sign is included in the equation to give a positive value of K .

In S.I. units it is expressed as $\rightarrow \text{N/m}^2$

Metric system units $\rightarrow \text{kg(F)}/\text{cm}^2$ or $\text{kg(F)}/\text{m}^2$

English system units $\rightarrow \text{lb(F)}/\text{in}^2$ or $\text{lb(F)}/\text{ft}^2$

Surface Tension & Capillarity

Due to molecular attraction, liquids possess certain properties such as cohesion and adhesion.

- Cohesion means inter-molecular attraction between molecules of the same liquid.
- Adhesion means attraction between the molecules of a liquid and the molecules of a solid boundary surface in contact with the liquid.

Surface tension is due to cohesion between liquid molecules at the surface.

Capillarity is due to both cohesion and adhesion

Surface Tension



A liquid molecule on the interior of the liquid body has other molecules on all sides of it, so that the forces of attraction are in equilibrium and the molecule is equally attracted on all the sides, as a molecule at point A shown in figure.

On the other hand, a liquid molecule at the surface of the liquid as at point B, does not have any liquid molecule above it, and consequently there is a net downward force on the molecule due to the attraction of the molecules below it.

This force on the molecules at the liquid surface is normal to the liquid surface. Apparently, owing to the attraction of liquid molecules below the surface, a film or a special layer seems to form on the liquid at the surface, which is in tension, and small bodies can be supported over it.

For example, a small needle placed quantity upon the water surface will not sink but will be supported by the tension at the water surface. The property of the liquid surface film to exert a tension is called the surface tension. It is denoted by σ (sigma) and it is the force required to maintain unit length of the film in equilibrium.

- In S.I. units $\rightarrow$ N/m
- Metric gravitational units $\rightarrow$ kg(F)/cm or kg(F)/m
- English gravitational units $\rightarrow$ lb(F)/ in or lb(F)/ft

As surface tension is directly dependent upon inter-molecular cohesive forces, its magnitude for all liquids decreases as the temperature rises.

(i) Pressure intensity inside a droplet

Consider a spherical droplet of radius ' r ' having internal pressure intensity ' p ' in excess of outside pressure intensity. If the droplet is cut into two halves, then the forces acting on one half will be those due to Pressure intensity ' p ' on the projected area (πr^2) and the Tensile force due to surface tension ' σ ' acting around the circumference ($2 \pi r$)

These two forces will be equal and opposite for equilibrium:

$$\begin{aligned} p(\pi r^2) &= 2\sigma(\pi r) \\ p &= \frac{2\sigma}{r} \end{aligned}$$

From the above equation, the internal pressure intensity increases with the decrease in the size of droplets.

(ii) Pressure intensity inside a soap bubble

A spherical soap bubble has two surfaces in contact with air - one inside and the other outside - each of which contributes the same amount of tensile force due to surface tension. As such, on a hemispherical section of a soap bubble of radius ' r ', the tensile force due to surface tension is equal to:

$$2\sigma(2\pi r)$$

However, the pressure forces acting on the hemispherical section of the soap bubble is the same as in the case of a droplet and is equal to $p(\pi r^2)$. Thus, equating the two forces for equilibrium, we have:

$$\begin{aligned} p(\pi r^2) &= 2\sigma(2\pi r) \\ p &= \frac{4\sigma}{r} \end{aligned}$$

(iii) Pressure intensity inside a liquid jet

Consider a jet of liquid of radius ' r ', length ' l ', and having internal pressure intensity ' p ' in excess of the outside pressure intensity. If the jet is cut into halves, then the forces acting on one half will be those due to Pressure intensity ' p ' on the projected area ($2\pi r$)

The Tensile force due to surface tension ' σ ' acting along the two sides ($2l$). These two forces will be equal and opposite for equilibrium and hence:

$$\begin{aligned} p(2rl) &= \sigma(2l) \\ p &= \frac{2\sigma}{r} \end{aligned}$$

Capillarity

If the liquid has greater adhesion than cohesion, then it will wet a solid surface with which it is in contact and will tend to rise at the point of contact, with the result that the liquid surface is concave upwards and the angle of contact θ is less than 10° .

For example, if a glass tube of small diameter is partially immersed in water, the water will wet the surface of the tube and it will rise in the tube to some height above the normal water surface, with the angle of contact θ being zero. The wetting of solid boundary by

Liquid results in creating pressure decrease within the liquid and hence the rise in the liquid surface takes place, so that the pressure within the column at the elevation of the Surrounding liquid surface is the same as the pressure at this elevation outside the column.

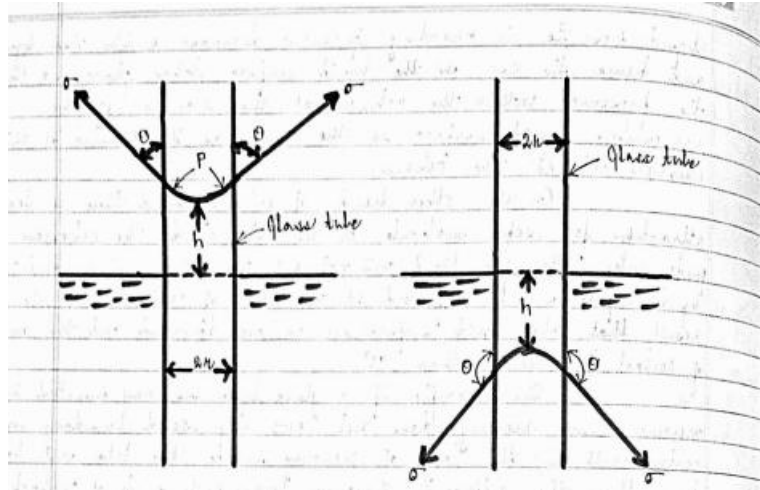
On the other hand, if for any liquid there is less attraction for solid molecules or in other words the cohesion predominates, then the liquid will not wet the solid surface and the liquid surface will be depressed at the point of contact, with the result that the liquid surface is concave downward and the angle of contact θ is greater than 90° .

For example, if a glass tube is now inserted in mercury, since mercury does not wet the solid boundary in contact with it, the level of mercury inside the tube will be lower than the adjacent mercury level, with angle of contact θ equal to about 130° .

The tendency of the liquid which do not adhere to the solid surface, resulting in creating an increase of pressure on the liquid surface, is to become of the internal pressure increased. The elevation of the meniscus in the tube is lowered to the level where the pressure is the same as that in the surrounding liquid.

Such a phenomenon of rise or fall of liquid surface relative to the adjacent general level of liquid is known as capillarity. Accordingly, the rise of liquid surface is designated as capillary rise and lowering of liquid surface as capillary depression.

- S.I. units $\rightarrow$ N or mm
- Metric system $\rightarrow$ cm or mm
- English system $\rightarrow$ inch or ft



The capillary rise (or depression) can be determined by considering the conditions of equilibrium in a circular tube of small diameter inserted in a liquid. It is supposed that the level of liquid has risen (or fallen) by ' h ' above (or below) the general liquid surface when a tube of radius ' r ' is inserted in the liquid.

For the equilibrium of vertical forces acting on the mass of liquid lying above (or below) the general liquid level, the weight of liquid column ' h ' must be balanced by the force at surface of the liquid due to surface tension σ . Thus, equating the two forces, we have:

$$h = \frac{2\sigma \cos \theta}{SWr}$$

As stated earlier, the contact angle θ for water and glass is equal to zero. Thus the value of $\cos \theta$ is equal to unity and hence ' W ' is given by the expression ...

$$h = \frac{2\sigma}{Wr}$$

Above equation for capillary rise (or depression) indicates that the smaller the radius ' r ' the greater is the capillary rise (or depression).

If a tube of radius ' r ' is inserted in mercury (*sp. gr.* S_2) above which a liquid of (*sp. gr.* S_1) lies, then by considering the conditions of equilibrium it can be shown that the capillary depression ' h ' is given by:

$$h = \frac{2\sigma \cos \theta}{r w (S_1 - S_2)}$$

Further, if two vertical parallel plates 't' distance apart and each of width 'l' are held partially immersed in a liquid of surface tension ' σ ' and sp. gr. ' S ', then capillary rise (or depression) ' h ' may be determined by equating the weight of the liquid column h (or the total internal pressure in the case of capillary depression) ($S r h l t$) to the force due to surface tension ($2\sigma \cdot l \cdot \cos \theta$).

Thus we have:

$$\sigma r h l t = 2\sigma r \cos \theta$$

or

$$h = \frac{2\sigma \cos \theta}{\sigma w r}$$

Fluid Pressure and its Measurement

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Fluid Pressure at a Point

Pressure or intensity of pressure may be defined as the force exerted on a unit area.

If **F** represents the total force uniformly distributed over an area **A**, the pressure at any point is:

$$p = \frac{F}{A}$$

However, if the force is not uniformly distributed, the expression will give the average value only.

When pressure varies from point to point on an area, the magnitude of pressure at any point can be obtained:

$$p = \frac{dF}{dA}$$

Where dF represents the force acting on an infinitesimal area dA .

- In S.I. units, pressure is $\rightarrow \text{N/m}^2$

- Metric gravitational units $\rightarrow \text{kg(f)}/\text{cm}^2$ or $\text{kg(f)}/\text{m}^2$

ariation of Pressure in a Fluid

Consider a small fluid element of size $\delta x \times \delta y \times \delta z$ at any point in a static mass of fluid. Since the fluid is at rest, the element is in equilibrium under the forces acting on it.

The forces acting on the element are the pressure forces on its faces and the self-weight of the element.

Thus, the pressure intensity on the left-hand face of the element is:

$$\left[p - \left(\frac{\partial p}{\partial x} \right) \frac{\delta x}{2} \right]$$

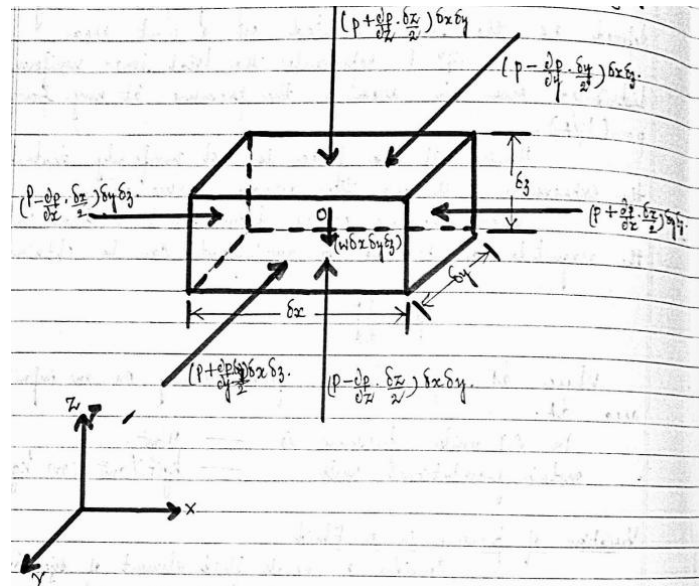
and the pressure intensity on the right-hand face of the element is:

$$\left[p + \left(\frac{\partial p}{\partial x} \right) \frac{\delta x}{2} \right]$$

The corresponding pressure forces on the left-hand and the

Right hand faces of the element are

$$\left[p - \left(\frac{\partial p}{\partial x} \right) \frac{\delta x}{2} \right] \delta y \delta z \text{ and } \left[p + \left(\frac{\partial p}{\partial x} \right) \frac{\delta x}{2} \right] \delta y \delta z$$



Respectively, likewise the pressure intensities and the corresponding pressure forces on the other faces of the element may be obtained and is shown in figure.

Further, if w is the specific weight of the fluid, then the weight of the element acting vertically downwards is:

$$(w \cdot \delta x \cdot \delta y \cdot \delta z)$$

Since the element is in equilibrium under these forces, the algebraic sum of the forces acting on it in any direction must be zero. Thus, considering the forces acting on the element along x, y, and z axes the

Following equations are obtained:

$$\begin{aligned} \Sigma F_x &= 0 \quad \text{or} \\ \Rightarrow \left(p - \frac{\partial p}{\partial x} \cdot \frac{\delta x}{2} \right) \delta y \delta z - \left(p + \frac{\partial p}{\partial x} \cdot \frac{\delta x}{2} \right) \delta y \delta z &= 0 \\ \frac{\partial p}{\partial x} &= 0 \dots (1) \end{aligned}$$

$$\begin{aligned} \Sigma F_y &= 0 \quad \text{or} \\ \Rightarrow \left(p - \frac{\partial p}{\partial y} \cdot \frac{\delta y}{2} \right) \delta x \delta z - \left(p + \frac{\partial p}{\partial y} \cdot \frac{\delta y}{2} \right) \delta x \delta z &= 0 \\ \frac{\partial p}{\partial y} &= 0 \dots (2) \end{aligned}$$

$$\begin{aligned} \Sigma F_z &= 0 \quad \text{or} \\ \Rightarrow \left(p - \frac{\partial p}{\partial z} \cdot \frac{\delta z}{2} \right) \delta x \delta y - \left(p + \frac{\partial p}{\partial z} \cdot \frac{\delta z}{2} \right) \delta x \delta y - (w \cdot \delta x \delta y \delta z) &= 0 \\ \frac{\partial p}{\partial z} &= -w \dots (3) \end{aligned}$$

Equations (1), (2), and (3) indicate that the pressure intensity p at any point in a static mass of fluid does not vary in x and y directions and it varies only in z-direction.

Hence, the partial derivative in eq (3) may be reduced to total derivative as:

$$\frac{dp}{dz} = -w \dots (4)$$

In vector notation, eq (4) may be expressed as:

$$-\text{grad } p = w \cdot k = \rho g k$$

where k is unit vector parallel to z -axis.

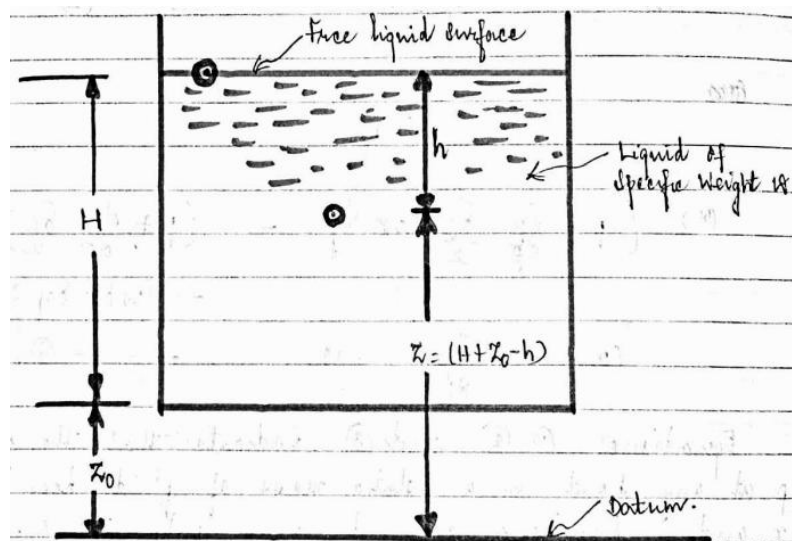
The minus sign indicates pressure decreases in the direction in which z increases, i.e., upward direction.

Pressure at a Point in a Liquid

A liquid may be considered as incompressible fluid for which w is constant, and hence integration of equation (4) gives

$$p = -wz + C \dots (5)$$

in which p is the pressure at any point at an elevation z in the static mass of liquid and C is the constant of integration. Liquids have a free surface at which the pressure of atmosphere acts.



Thus as shown in figure for a point lying in the free surface of the liquid

$$z = (H + Z_0)$$

and if p_a is the atmospheric pressure at the liquid surface then, from equation (5), the constant of integration

$$C = [p_a + w(H + Z_0)]$$

Substituting this value of C in eq. (5) it becomes

$$p = -wz + [p_a + w(H + Z_0)] \dots (6)$$

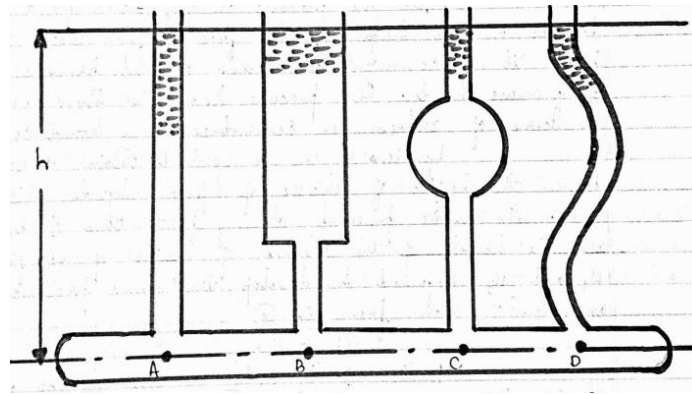
Now if a point is lying in the liquid mass at a vertical depth h below the free surface of the liquid, then as shown in figure, for this point

$$z = (H + Z_0 - h)$$

and from eq. (6)

$$p = p_a + wh \dots (7)$$

It is evident from equation (7) that the pressure at any point in a static mass of liquid depends upon the vertical depth of the point below the free surface and the specific weight of the liquid, and it does not depend upon the shape and size of the bounding container.



In the above figure although the containers of different shapes are interconnected, so that the total weight of the liquid in each part differs, the pressures at points A, B, C and D lying on the same horizontal level and at the same vertical depth ' h ' below the free surface of the liquid will be the same. Since the atmospheric pressure at a place is constant, at any point in a static mass of liquid often only the pressure in excess of the atmospheric pressure is considered, in which case eq. (7) becomes

$$p = wh \dots (8)$$

Pressure Head

The vertical height or free surface above any point in a liquid at rest is known as pressure head.

From eq. (8) the pressure head may be expressed as

$$h = \frac{p}{w} \dots (9)$$

Since the pressure at any point in a liquid depends on the height of the free surface above the point, it is convenient to express a fluid pressure in terms of pressure head. The pressure head is then expressed in terms of meters (or centimeters) of a liquid column.

Equation (8) can be used to obtain a relationship between the heights of columns of different liquids which would develop the same pressure at any point. Thus if h_1 and h_2 are the heights of columns of liquids of specific weights w_1 and w_2 required to develop the same pressure p at any point, then from eq. (8):

$$S_1 h_1 = S_2 h_2$$

Further, if h_1 and h_2 are the depths of two points below the free surface in static mass of liquid of specific weight w , and p_1 & p_2 are the respective pressure intensities at these points, then from eq. (8) the pressure difference between these points is obtained as:

$$(p_1 - p_2) = w(h_1 - h_2)$$

Thus it may be stated that the differences in pressures at any points in a static mass of liquid varies directly as the difference in depth (or elevation) of the two points.

Isothermal State

If it is assumed that the variation of p and ρ is according to isothermal condition, i.e., the temperature of atmosphere is assumed to have a constant value (which may however be true only over a relatively small vertical distance), then according to Boyle's law:

$$\frac{p}{\rho} = \frac{p_0}{\rho_0}$$

where p_0 and ρ_0 are the values of the pressure and density of the gas at initial condition at some reference level (for example, at the earth's surface, i.e., at $z = 0$).

Adiabatic State

It is well known fact that the temperature in the atmosphere decreases rapidly as one goes to higher altitudes. Thus it is assumed that no heat is

added or taken away from a certain column of air, then this column is said to be under adiabatic conditions.

For an adiabatic condition:

$$\frac{p}{\rho^k} = \frac{p_0}{\rho_0^k} \text{ (or) } \rho = \rho_0 \left(\frac{p}{p_0} \right)^{\frac{1}{k}}$$

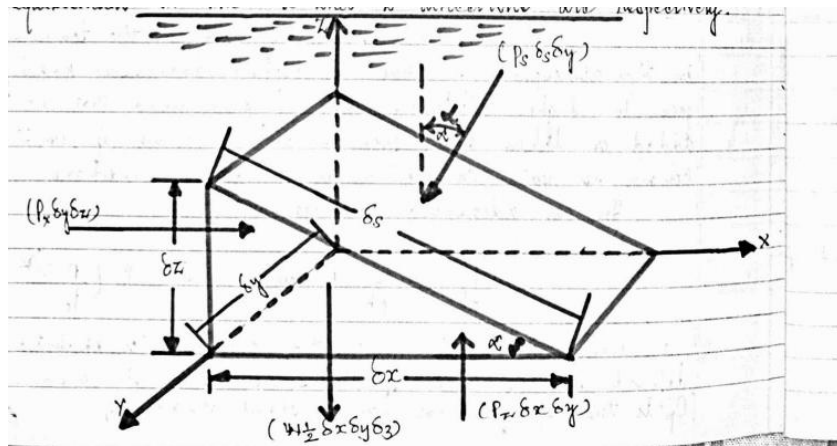
in which k is the adiabatic exponent (or adiabatic index), defined as the ratio of the specific heat at constant pressure C_p to the specific heat at constant volume C_v .

Pressure The Same in All Directions – Pascal's Law

The pressure at any point in a fluid at rest has the same magnitude in all directions.

In other words, when a certain pressure is applied at any point in a fluid at rest, the pressure is equally transmitted in all the directions and to every other point in the fluid. This fact was established by B. Pascal, a French Mathematician, in 1653, and accordingly it is known as Pascal's Law.

Let us consider an infinitesimal wedge-shaped element of fluid at rest as a free body. The element is arbitrarily chosen and has the dimensions as shown in figure. Since in a fluid at rest there can be no shear forces, the only forces acting on the free body are the normal pressure forces exerted by the surrounding fluid on the plane surfaces, and the weight of the element. As the element is in equilibrium, the sum of the forces components on the element in any direction must be equal to zero. So, the equations of equilibrium in the X and Z directions are respectively...



$$P_x \delta y \delta z - P_s \delta s \delta y \sin \alpha = 0$$

$$P_z \delta x \delta y - P_s \delta s \delta y \cos \alpha - w \frac{1}{2} \delta x \delta y \delta z = 0$$

in which P_x, P_z, P_s are the average pressures on the three faces and w is the specific weight of the fluid.

Since $\delta z = \delta s \sin \alpha$ and $\delta x = \delta s \cos \alpha$, the above equations simplify to

$$P_x \delta y \delta z - P_s \delta y \delta z = 0$$

$$P_z \delta x \delta y - P_s \delta x \delta y - \frac{1}{2} w \delta x \delta y \delta z = 0$$

$$\delta_s$$

The third term of the second equation is much smaller than the other two terms and hence it may be neglected. Now by dividing the equations by.

$\delta y \delta z$ and $\delta x \delta y$ respectively and taking the limit, so that the element is reduced to a point, it follows from the equations that $P_s = P_x = P_z$.

Since angle α is chosen arbitrarily, this equation proves that the pressure is the same in all directions at a point in a static fluid.

Thus Pascal's law is made use of in the construction of machines such as hydraulic press, hydraulic jack, hydraulic lift, hydraulic crane, etc., in which by the application of relatively small forces considerably larger forces are developed.

Atmospheric, Absolute Gage and vacuum pressure

The atmospheric air exerts a normal pressure upon all surface with which it is in contact, and is known as atmospheric pressure. The atmospheric pressure varies with the attitude and it can be measured by

means of a barometer. As such it is also called the barometer pressure. At sea level under normal conditions the equivalent values of the atmospheric pressure are

$10.1043 \times 10^4 \text{ N/m}^2$; (or) $1.03 \text{ kg(f)}/\text{cm}^2$; (or)
 10.3 m of water; or 16 cm of mercury.

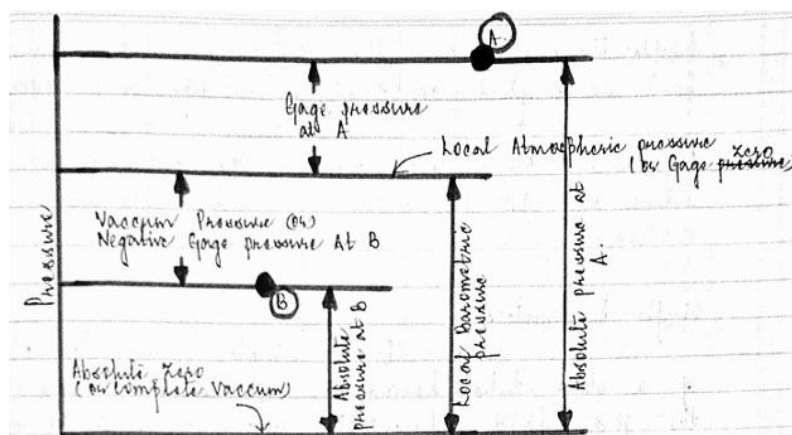
Fluid pressure may be measured with respect to any arbitrary datum. The two most common datum used are

1. Absolute zero pressure
2. Local atmospheric pressure

When pressure is measured above absolute zero (or complete vacuum,) it is called a absolute pressure.

When it is measured either above or below atmospheric pressure as datum, it is called gage pressure. If the pressure of a fluid is below atmospheric pressure it is designated as vacuum pressure Or section pressure on negative gage pressure), and its gage value is the amount by which it is below that of the atmospheric pressure. A gage which measures vacuum pressure is known as vacuum gage.

All values of absolute pressure are positive, since in the case of fluid the lowest absolute pressure which can possibly exists corresponding to absolute zero or complete vacuum. However, gage pressure are positive if they are above that of the atmosphere and negative if they are vacuum pressures.



Relationship between absolute, gage and vacuum pressures

Absolute pressure = Atmospheric pressure + Gage Pressure

Absolute Pressure = Atmospheric pressure - vacuum pressure

Measurement of pressure

The various devices adopted for measuring

Fluid pressure

- Manometer
- Mechanical gage.

Manometers are those pressure measuring devices which are based on the principle of balancing the column of liquid by the same or another column of liquid. Manometers are

- Simple manometer
- Differential manometer

Simple manometers are those which measure pressure at a point in a fluid contained in a pipe or a vessel.

Differential Manometers measure the difference of pressure between any two points in a fluid contained in a pipe or vessel.

Simple Manometers:

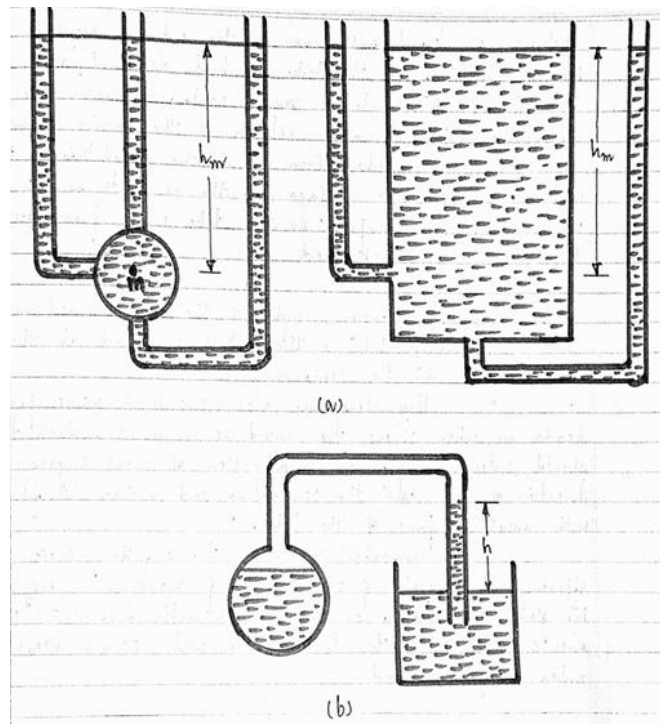
In general a simple manometer consists of a glass tube having one of its ends connected to the gage point where the pressure is to be measured and the other remains open to atmosphere. They are

- Piezometers
- u-tube manometers
- single column manometers

Piezometers:

A Piezometer is the simplest form of manometer which can be used for measuring moderate pressure of liquids. It consists of a glass tube inserted in the wall of a pipe or a vessel, containing a liquid whose pressure is to be measured. The tube extends vertically upward to such a height that liquid can freely rise in it without overflowing. The pressure at any point in the liquid is indicated by the height of the liquid in the tube above that point, which can be read on the scale attached to it. Thus, if ω is the specific weight of the liquid, this pressure at point m in figure(a) is $P_m = \omega h_m$. In other words, h_m is the pressure head at m. Piezometers measure gage pressure.

only, since the surface of the liquid in the tube is subjected to atmospheric pressure.



Piezometer may be inserted either in the top, or the side, or the bottom of the container, but the liquid will rise to the same level in the three tube.

Negative gage pressures (or pressures less than atmospheric) can be measured by means of the piezometer shown in the fig(b). It is evident that if the pressure in the container is less than atmospheric no column of liquid will rise in the ordinary piezometer. But if the top of the tube is bent downward and its lower end dipped into a vessel containing water. The atmosphere will cause a column of the liquid to rise a height 'h' in the tube, from which the magnitude of the pressure on the free surface in the container is the same as that at free surface in the tube which from equation $p = wh$ may be expressed as

$$P = -wh$$

Where 'w' is the specific weight of the liquid used in the vessel, conversely - h is the pressure head at the free surface in the container.

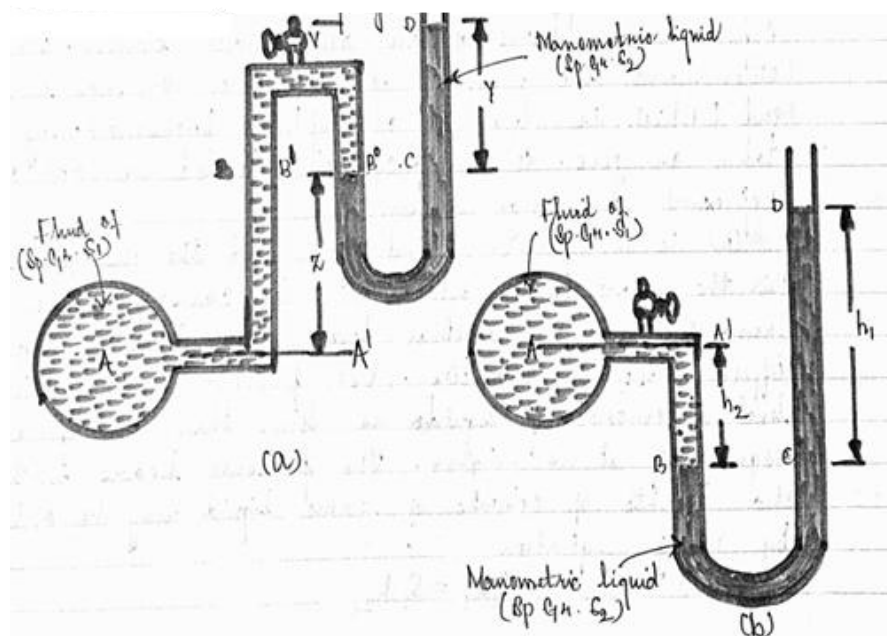
Piezometers are also used to measure pressure heads in pipes where the liquid is in motion. Such tubes should enter the pipe in a direction at right angles to the direction of flow and the connecting should be fluid with inner surface of the pipe.

In order to prove the capillary motion from affecting the height of the column of liquid in a piezometer the glass tube having an internal diameter less than 12mm should not be used. For precise work 25mm diameter tubes may be used.

U-tube Manometer

Piezometers cannot be used when large piezometer in the light liquids are to be measured. Since this would require very long tubes, which cannot be handled conveniently. Furthermore gas pressure cannot be measured by means of piezometer because a gas forms no free atmosphere surface.

A U-tube Manometer consists of a glass tube bent in U-shape, one end which is connected to the gage point and the other end remains open to the atmosphere. The tube contains a liquid of specific gravity greater than that of the fluid of which the pressure is to be measured. Sometimes more than one liquid may also be used in the manometer. The choice of the manometer liquid, however, depends on the range of pressure to be measured. For low pressure range, liquids of lower specific gravities are used and for high pressure range generally mercury is employed.



Where one of the limbs of the U-tube manometer is connected to the gage point, the fluid from the container or pipe will enter the connected limb of the manometer thereby causing the manometric liquid to rise from either 'A' or from the free surface in the open end of the manometer and write the

pressure there in an appropriate units (say, meters of water or other fluid, or N/m^2 (or), kg(f)/cm^2 , or kg(f)/m^2).

Of The pressure is unknown at A. It may be expressed in terms of an appropriate symbol. On the other hand, the pressure at the free surface in the open end (which is equal to atmospheric pressure) may be taken as zero, so that the equation formed in each case will represent the gauge pressure.

(2) In the pattern found above, add the change in pressure in the same units which will be caused while proceeding from one level to another level of contact of liquid of different specific gravities. Use positive sign if the next level of contact is lower than the first, and negative if it is higher. The pressure heads on terms of the heights of columns of same liquid may be obtained by using the equation:

$$S_1 h_1 = S_2 h_2$$

(3) Continue the process as in (2) until the other end of the gauge is reached and equate the expression to the pressure at that point, known or unknown.

The expression will contain only one unknown viz., the pressure at A, which may then be evaluated thus from manometer arrangement. Shown in above figure, the gauge equation may be written as:

Starting from 'A', if P_A is the unknown pressure intensity at A, W is the specific weight of water and S_1 is the specific gravity of the liquid in the container, then the pressure head at A = $\left(\frac{P_A}{W S_1}\right)$ in terms of liquid at A. Since all points lying at the same horizontal level in the same continuous static mass of liquid have the same pressure:

$$\text{Pressure at A} = \text{Pressure at A'}$$

From A' to B' there being an increase in elevation, pressure head decreases, so that pressure head at B' = $\left(\frac{P_A}{W S_1} - z\right)$

Again from the above consideration:

$$\text{Pressure at B'} = \text{Pressure at B} = \text{Pressure at C}.$$

From C to D there being increase in elevation, pressure head decreases. If S_2 represents the specific gravity of the manometric liquid, then from the

equation $S_1 h_1 = S_2 h_2$ the pressure head, in terms of liquid at A, equivalent to the column $CD (= y)$, of the manometric fluid $= y(S_2/S_1)$. Thus, pressure head at

$$D := \left(\frac{P_A}{wS_1} \right) - z - y \left(\frac{S_2}{S_1} \right).$$

But at D there

being atmospheric pressure, the pressure head $= 0$, in terms of the gauge pressure. Thus, equating the pressure head at D, the gauge equation becomes:

$$\begin{aligned} \frac{PA}{WS_1} - z - \frac{yS_2}{S_1} &= 0 \\ \frac{P_A}{wS_1} &= z + \frac{yS_2}{S_1} \quad \text{---(1)} \end{aligned}$$

Equation (1) represents the pressure heads in terms of the liquid at A. however if the pressure heads are expressed in terms of water the following equation is obtained.

$$\frac{P_A}{w} = zS_1 + yS_2$$

If A contains gas, its specific weight being quite small, the specific gravity S_1 of the gas (with respect to water) will be so small that zS_1 may be neglected. In which case the pressure head of the gas at A in terms of water is given by.

$$\frac{PA}{WS_1} = yS_2$$

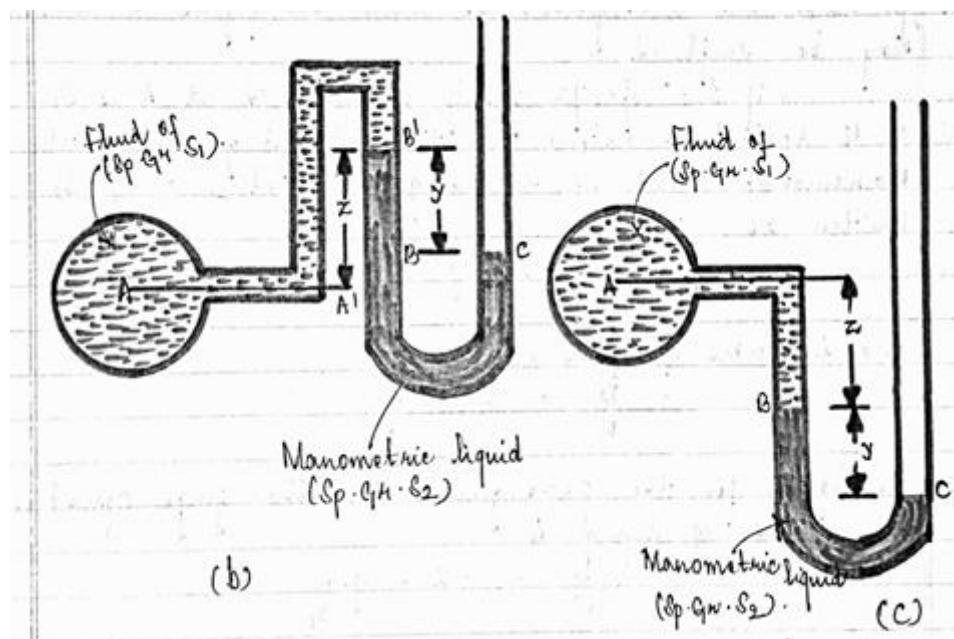
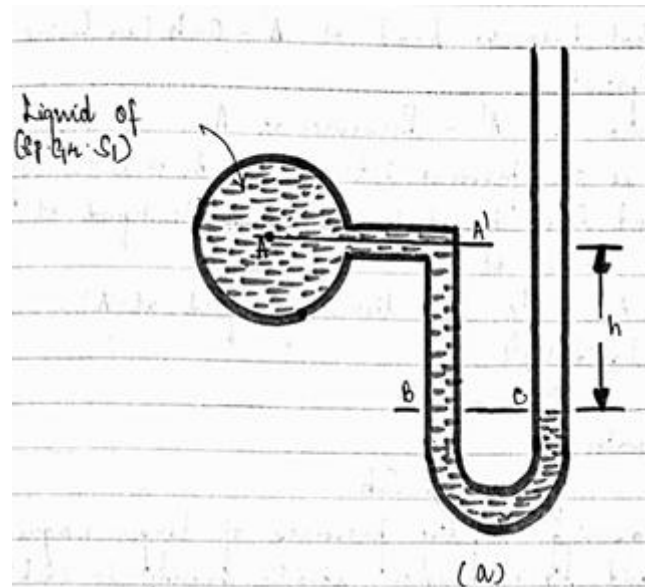
Fig(b) shows another arrangement for measuring pressure at A by means of a U-tube manometer. The equation in terms of A

$$\frac{P_A}{wS_1} = h_1 \frac{S_2}{S_1} - h_2$$

And in terms of water as

$$\frac{P_A}{w} = h_1 S_1 + h_2 S_1$$

A U-tube manometer can also be used to measure negative or vacuum pressure. For measurement of small negative pressure, a U-tube manometer without any manometric liquid may be used. It is evident that for pressure at A being negative (i.e., less than atmospheric pressure) the liquid surface in the open limb of the manometer will be below A. the pressure at A may be determined from the gage equation as:



At C the pressure head = 0, in terms of gage pressure. Further, pressure at C = Pressure at B. From B to A', there being increase in elevation, the pressure head decreases, so that pressure head at A' = 0 - h (in terms of liquid at A). Again

Pressure at A = Pressure at A'

Now, if P_A is the pressure intensity at A , W is the specific weight of water and S_1 is the specific gravity of the liquid at A , then the pressure head at

$$A = (P_A / WS_1) \text{ (in terms of liquid at } A \text{)}.$$

Thus, in terms of liquid at A :

$$\frac{P_A}{WS_1} = h$$

And in terms of water:

$$\frac{P_A}{W} = S_1 h$$

For measuring negative pressures of larger magnitude, a manometric liquid having higher specific gravity is employed, for which the arrangement as shown in figures (B) and (C) may be employed.

If the specific gravity of the fluid at A is S_1 (with respect to water) and the specific gravity of the manometric liquid is S_2 , the gauge equation may be written as:

$$\frac{P_A}{WS_1} = z - y \cdot \frac{S_2}{S_1}$$

And in terms of water as:

$$\frac{P_A}{W} = zS_1 - yS_2$$

Similarly, for the arrangement (C), the gauge equation in terms of liquid A :

$$\frac{P_A}{WS_1} = -z - y \cdot \frac{S_2}{S_1}$$

And in terms of water as

$$\frac{P_A}{W} = -zS_1 - yS_2$$

Single Column Manometer

The U-tube manometer described above usually require readings of fluid levels at two or more points, since a change in pressure comes a rise of

liquid in one limb of the manometer and a drop in the other. This difficulty may however be overcome by using single column manometer. A single column manometer is a modified form of a U-tube manometer in which a shallow reservoir having a cross-sectional area (about 100 times) as compared to the area of the tube is introduced into one limb of the manometer as shown in figure. For any variation in pressure, the change in the liquid level in the reservoir will be so small that it may be neglected, and the pressure is indicated approximately by the height of the liquid in the other limb. As such only one reading in the narrow limb of the manometer need be taken for all pressure measurements. The narrow limb of the manometer may be vertical as in figure(a) or it may be inclined as in figure(b). the inclined type is useful for the measurement of small pressure.

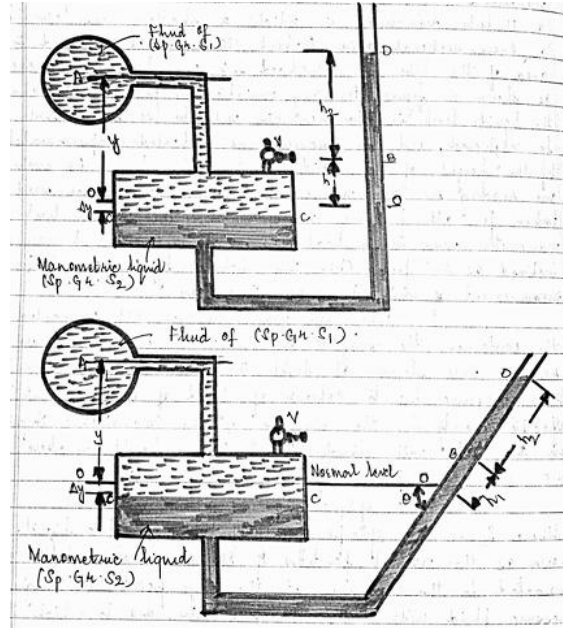
When the manometer is not connected to the container, the surface of the manometer liquid in the reservoir will stand at level 0-0, and since it is subjected to a pressure due to column of fluid of height 'y' and specific gravity S_1 , the surface of the manometric liquid in the tube will stand at B, at a height h_1 above 0-0 such that from

$$S_1 h_1 = S_2 h_2$$

$$y S_1 = h_1 S_2 \text{ --- (1)}$$

When S_2 is the specific gravity of the manometric liquid. This is known as the normal position of the manometric liquid. on being connected to the container at the gage point, the high pressure fluid will enter the reservoir, due to which there will be drop in the manometric liquid surface in the reservoir by a distance ' Δy ' and a consequent rise in the tube by a distance h_2 above B. if 'A' and 'a' are the cross-sectional areas of the reservoir and the tube respectively

$$A(\Delta y) = a h_2 \text{ --- (2)}$$



If P_A is the pressure intensity at A and W is the specific weight of water, then starting from D the following gage equation in terms of water is obtained

$$0 + (h_2 + h_1 + \Delta y)s_2 - (\Delta y)s_1 - ys_1 = \frac{p_A}{W}$$

Introducing equation (1) and (2) in equation (2) it becomes

$$\frac{P_A}{W} = h_2 \left[S_2 + (S_2 - S_1) \frac{q}{A} \right]$$

By making the reservoir sufficiently large, the ratio $\frac{a}{A}$ can be made so small that Δy is negligible and height $(h_2 + h_s)$ is a measure of the pressure head at the contact level C – C in the reservoir. In which case equation (4) reduces to

$$\frac{p_A}{W} = h_2 S_2$$

So that only one reading of the height of level of liquid in the narrow tube is required to be taken to obtain the pressure head at A. However, if Δy is appreciable, then since the term within the brackets of the right side of equation (4) are constant, the scale on which h_2 is read can be so graduated as to correct for Δy so that again only one reading of the height of liquid level in the narrow tube is required to be taken, which will directly give the pressure head at A.

A single tube manometer can be made more sensitive by making its narrow tube inclined as shown in figure. With this modification the distance moved by the liquid in the narrow tube shall be comparatively more, even for small pressure intensity at A. As Before when the manometer is not connected to the container, the manometric liquid surface in the reservoir will stand at O – O and that in the tube will stand at B , so that

$$yS_1 = (h_1 \sin \theta)S_2 \text{ --- (5)}$$

Due to high pressure fluid entering the reservoir, the manometric liquid surface will shift to level C – C by a distance Δy and it will travel a distance BD equal to h_2 in the narrow tube. Thus,

$$A(\Delta y) = ah_2 \text{ --- (6)}$$

Again, starting from D, the gage equation in this case becomes

$$0 + (h_1 + h_2)(\sin \theta)S_2 + \Delta yS_2 - \Delta yS_1 = yS_1 = \frac{p_A}{w} \text{ --- (7)}$$

Introducing equations (5) and (6) in eq (7), it becomes

$$\frac{p_A}{w} = h_2 \left[S_2 \sin \theta + (S_2 - S_1) \frac{q}{A} \right] \text{ --- (8)}$$

Again, if the ratio $\frac{q}{A}$ is negligible then equation (8) reduces to

$$\frac{p_A}{w} = (h_1 \sin \theta)S_2 \text{ --- (9)}$$

Single column manometers can also be employed to measure the negative gage pressures.

Differential Manometer

For measuring the difference of pressure between any two points in a pipeline or in two pipes or container, manometer consist of a bent glass tube, the two ends of which are connected to each of the two gage points between which the pressure difference is required to be measured examples are:

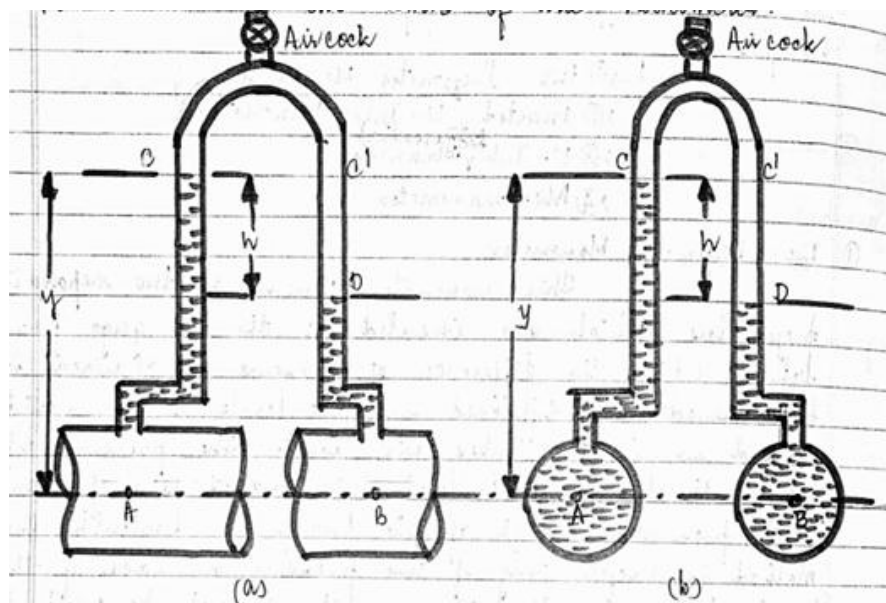
1. Two-piezometer manometer
2. Inverted U-tube manometer
3. U-tube Differentiate manometer
4. Micromanometer

(1) Two-Piezometer Manometer

This manometer consists of two separate piezometers which are inserted at the two gage points between which the difference of pressure is required to be measured. The difference in the levels of the liquid raised in the two tubes will denote the pressure difference between the two points. Evidently, this method is useful only if the pressure at each of the two points is small. Moreover, it cannot be used to measure the pressure difference in gases, for which the other types of differential manometers described may be used.

(2) Inverted U-Tube Manometer

It consists of a glass tube bent in U-shape and held inverted as shown in figure. Thus it is as if two piezometers described above are connected with each other at top, where two ends of the manometer are connected to the points between which the pressure difference is required to be measured, the liquid under pressure will enter the two limbs of the manometer thereby causing the air within the manometer to get compressed. The presence of the compressed air results in restricting the heights of the columns of liquids raised in the two limits of the manometer



An air is also provided at the top of the inverted U-tube facilities the raising of the liquid columns to suitable level in both the limbs by driving out a portion of the compressed air. It also permits the expression of air bubbles which might have been entrapped some where in the pipeline.

If P_A and P_B are the pressure intensities at point A and B between which the inverted U-tube manometer is connected. Then according to these pressure intensities, the liquid will rise above points A

and B up to C and D in the two limbs of the manometer. Now, if 'w' represents the specific weight of water and S_1 represents the specific gravity of the liquid at A and B , then commencing from A the pressure head at C in terms of water is equal to

$$\left(\frac{p_A}{w} - yS_1\right)$$

Since points C and C' are at the same horizontal level and in the same continuous static mass of fluid,

$$\text{Pressure at } C = \text{Pressure at } C'$$

Between points C' and D there is a column of compressed air. Since the specific weight of air is negligible as compared with that of liquid, the weight of air column between C' and D may be neglected. Hence,

$$\text{Pressure at } C' = \text{Pressure at } D$$

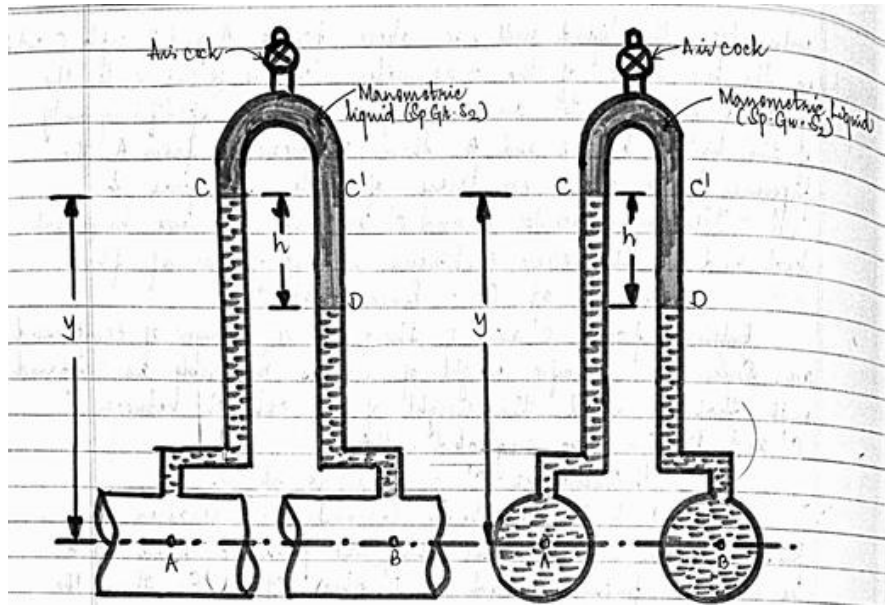
From D to B there being decrease in elevation, the pressure head increases, so that the pressure head at B is equal to pressure head at D plus $(y - h)S_1$. Thus the gage equation may be expressed as:

$$\frac{p_A}{w} + yS_1(y - h)S_1 = \frac{p_B}{w}$$

or,

$$\frac{p_A}{w} - \frac{p_B}{w} = hS_1 \text{ --- (1)}$$

Inverted U-tube manometers are suitable for the measurement of small pressure differences in liquids. Sometimes, instead of air, the upper part of this manometer is filled with a manometric liquid which is lighter than the liquid for which the pressure difference is to be measured and is immiscible with it. Use of manometric liquid increases the sensitivity of the manometers.



Again if P_A and P_B are the pressure intensities at points A and B between which the inverted U-tube manometer is collected, then corresponding to those pressure intensities the liquid will rise above points A and B upto C and D in the two limbs of the manometer as shown in figure now if 'w' represents the specific weight of water and S_1 and S_2 are the specific gravities of the liquid at A or B and the manometric liquid (in the upper part of the manometer) respectively then commencing from A, the pressure head at 'C' in terms of water is equal to

$$\left(\frac{P_A}{w} - yS_1 \right)$$

Since points C and C' are at the same horizontal level and in the same continuous static mass of liquid

$$\text{Pressure at C} = \text{pressure at C'}$$

From C' to D there being decrease in elevation, the pressure head increases, so that the pressure head at D in terms of water is equal to

$$\left[\left(\frac{P_A}{w} - yS_1 \right) + hS_2 \right]$$

Further, from D to B there is decrease in elevation and hence the gage equation becomes

$$\frac{P_A}{w} - yS_1 + hS_2 + (y - h)S_1 = \frac{P_B}{w}$$

or,

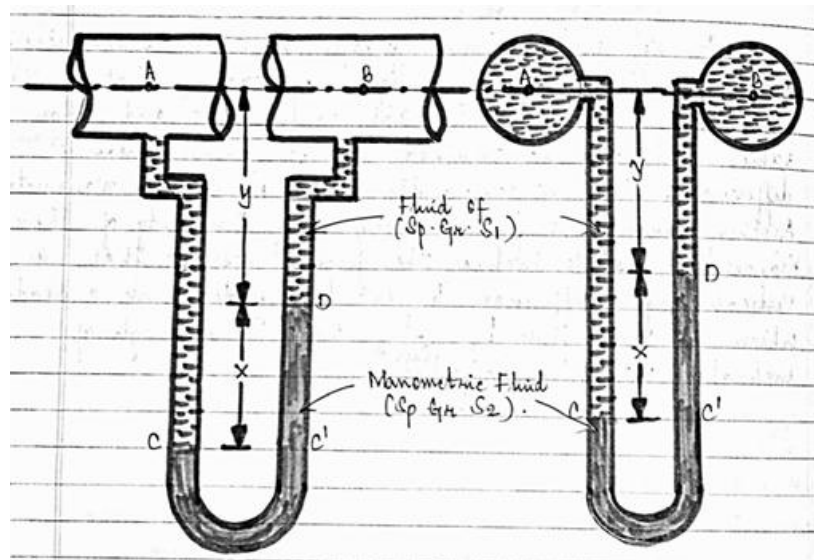
$$\frac{p_A}{w} - \frac{p_B}{w} = h(S_1 - S_2) \text{ --- (2)}$$

It is evident from equation (2) that as the specific gravity of the manometric liquid approaches that of the liquid at A and B ($S_1 - S_2$) approaches zero and large values of 'h' will be obtained even for small pressure differences, thus increasing the sensitivity of the manometer.

Another arrangement for increasing the sensitivity of these manometers is to incline the gage tubes so that a vertical gage difference 'h' is transported into a reading which is magnified by $\frac{1}{\sin \theta}$ where ' θ ' is angle of inclination with the horizontal.

U-Tube Differential Manometer

It consists of glass tube bent in U-shape, the two ends of which are connected to the two gage points between which the pressure difference is required to be measured below figure shows such an arrangement for measuring the pressure difference between any two points A and B. the lower part of the manometer contains a manometric liquid which is heavier than the liquid for which the pressure difference is to be measured and is immiscible with it.



When two limbs of the manometers are connected to the gage points A and B, then corresponding to the difference in the pressure intensities P_A and P_B the levels of manometric liquid in the two limbs of the manometer will be displaced through a distance 'X' as shown in figure by measuring

this difference in the levels of the manometric liquid, the pressure difference ($P_A - P_B$) may be computed as

if S_1 and S_2 are the specific gravities of the liquid at A and B, and the manometric liquid respectively. Then, by commencing at A where the pressure is p_A , the pressure head at C in terms of water is equal to

$$\left[\frac{p_A}{w} + (y + x)S_1 \right]$$

in which ' w ' is the specific weight of water. Further since points C and C' are at the same level and are lying in the same continuous static mass of liquid,

$$\text{Pressure at C} = \text{Pressure at C'}$$

Further, from C' to D there being an increase in elevation, the pressure head decreases, so that the pressure head at D in terms of water is equal to pressure difference, mercury

$$\left[\frac{p_A}{w} + (y + x)S_1 - xS_2 \right]$$

Similarly, from D to B there is an increase in elevation and hence the gage equation becomes:

$$\left[\frac{p_A}{w} + (y + x)S_1 - xS_2 - yS_1 \right] = \frac{p_B}{w}$$

or,

$$\frac{p_A}{w} - \frac{p_B}{w} = x(S_2 - S_1) \text{ --- (1)}$$

If, for example, the manometric liquid is mercury ($S_2 = 13.6$) and the liquid at A or B is water

($S_1 = 1$), then the difference in pressure heads at the points A and B is 12.6 times the deflection x of the manometric liquid in the two limbs of the manometer. As such the use of mercury as manometric liquid in U-tube manometers is suitable for measuring large pressure differences. However, for small pressure differences, mercury makes precise measurement difficult, and hence for such cases it is common to use a liquid which is only slightly heavier than the liquid for which the pressure difference is to be measured.

Often the points A and B between which the pressure difference is to be measured are not at the same level as shown in figure. For such cases also, by adopting the same procedure, the following gage equation may be obtained in order to compute the pressure difference between the points A and B :

$$\frac{p_A}{w} + (z + y + x)S_1 - xS_2 - yS_3 = \frac{p_B}{w}$$

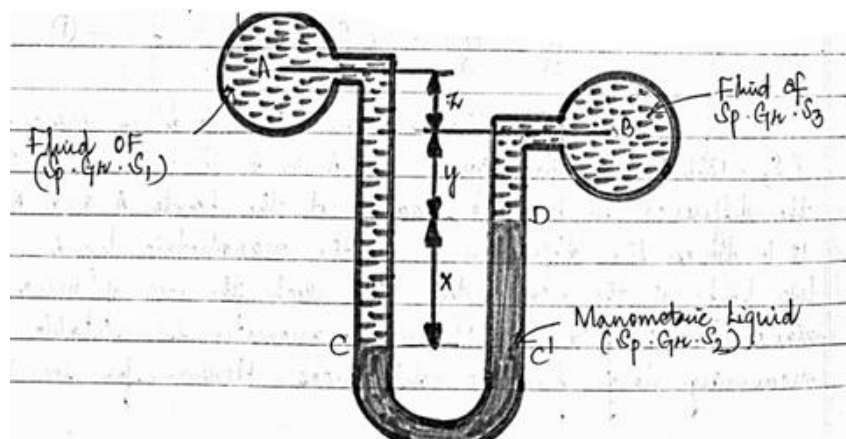
or,

$$\frac{p_A}{w} - \frac{p_B}{w} = [x(S_2 - S_1) + y(S_3 - S_1) - zS_1] \text{ --- (2)}$$

Equation (2) is, however, a general equation, which may be modified to derive the relations for different conditions. Thus, for example, if there is same liquid at A and B , then since $S_1 = S_3$, equation (2) becomes

$$\frac{p_A}{w} - \frac{p_B}{w} = [x(S_2 - S_1) - zS_1] \text{ --- (3)}$$

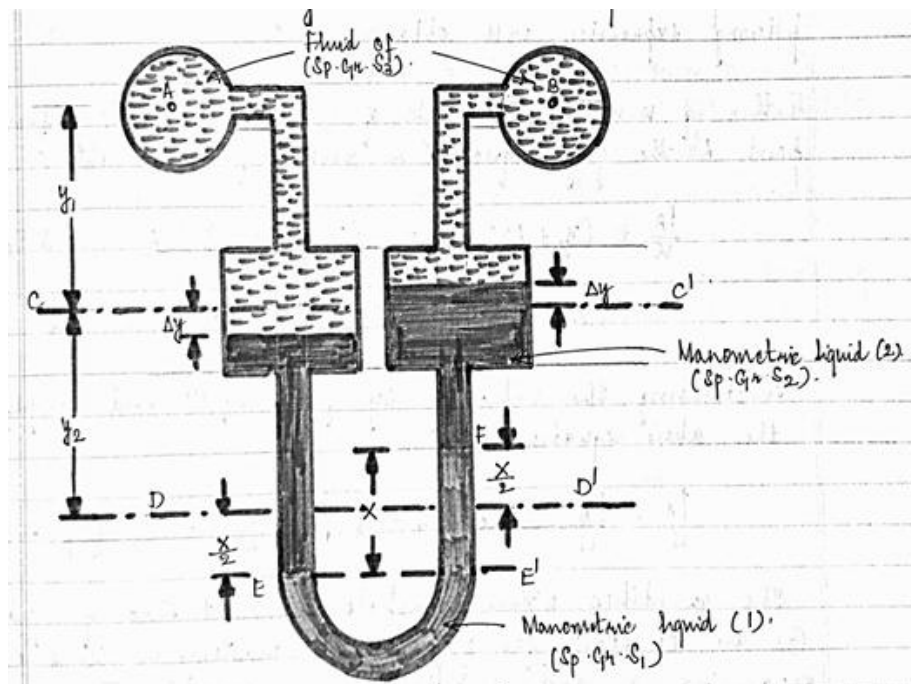
Further, if A and B are at the same level, then since $z = 0$, equation (3) becomes same as equation (1).



Micromanometer: -

For the measurements of very small pressure differences, or for the measurements of pressure difference with very high precision special forms of manometers called micromanometer are used of consists of glass U-tube provided with two transparent basins of wider sections at the top of the two limbs as shown in figure. The manometer contains two manometric liquids of different specific gravities and immiscible with each other and with the fluid for which the pressure difference is to be

measured. Before the manometer is connected to the pressure points A and B, both the limbs are subjected to the same pressure.



As such the heavier manometric liquid of Sp. gm S_1 will occupy the level DD' and Higher nanometric liquid Sp.gm of S_2 will occupy the level CC'. When the manometer is connected to the pressure points A and B where the pressure intensities are p_A and p_B respectively, such that $p_A > p_B$, then the level of the Highter manometric liquid will fall in the left basin and rise in the right basin by the same amount ' Δy '. Similarly, the level of the heavier manometric liquid will fall in the left limb to point E and rise in the right limb to point F .

If A and a' are the cross-sectional areas of the basin and the tube respectively, then since the volume of the liquid displaced in each basin is equal to the volume of the liquid displaced in each limb of the tube, the following expression will obtain:

$$A(\Delta y) = a\left(\frac{x}{2}\right) \text{ --- (1)}$$

Further, if 'w' is specific weight of water, then, starting from point A the gage equation in terms of water column:

$$\begin{aligned} \frac{p_A}{w} + (y_1 + \Delta y)S_3 + \left(y_2 - \Delta y + \frac{x}{2}\right)S_2 - xS_1 \\ - \left(y_2 - \frac{x}{2} + \Delta y\right)S_2 - (y_1 - \Delta y)S_3 = \frac{p_B}{w} \end{aligned}$$

Substituting the value of Δy from eq (1) and simplifying the above equation:

$$\frac{p_A}{w} - \frac{p_B}{w} = x \left[S_1 - S_2 \left(1 - \frac{a}{A} \right) - S_3 \frac{a}{A} \right] - - - (2)$$

The quantities within brackets on right side of equation (2) are constant for a particular manometer. Thus, by measuring x and substituting in equation (2), the pressure difference between any two points can be known.

If the cross-sectional area of the basin is large as compared with the cross-sectional area of the tube, then the ratio a/A is very small and the equation (2) reduces to

$$\frac{p_A}{w} - \frac{p_B}{w} = x(S_1 - S_2)$$

By selecting the two manometric liquids such that their specific gravities are very nearly equal, then a measurable value of x may be achieved even for a very small pressure difference between the two points.

Mechanical Gages

Mechanical gages are those pressure measuring devices which embody an elastic element, which deflects under the action of the applied pressure, and this movement, mechanically magnified, operates a pointer moving against a graduated circumferential scale.

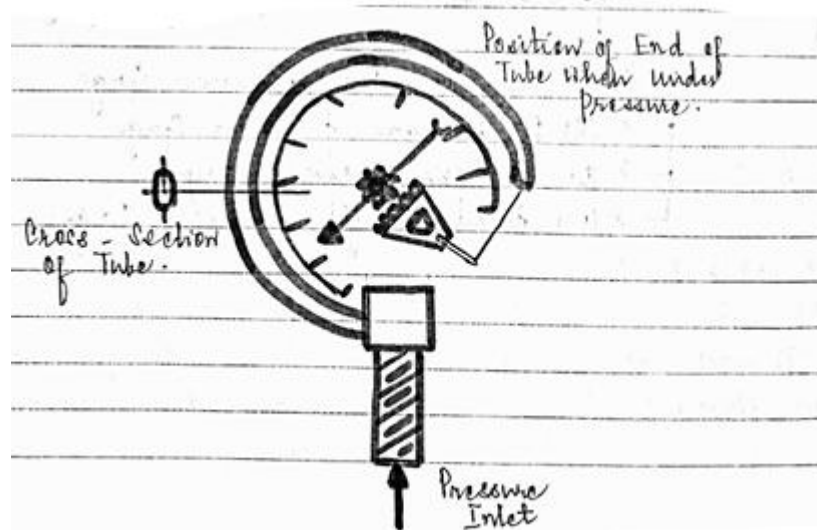
Generally these gages are used for measuring high pressures and where high precision is not required. Some of the mechanical pressure gages are:

1. Bourdon Tube Pressure Gage
2. Diaphragm Pressure Gage
3. Bellows Pressure Gage
4. Dead-weight Pressure Gage

1) Bourdon Tube Pressure Gage

It is the most common type of pressure gage which was invented by E. bourdon (1808-84). The pressure responsive elements in this gage is a tube of steel or bronze which is of ellipsis cross-section and is curved into a circular arc. The tube is free to move. The other end of the tube, through

which the fluid enters, is rigidly fixed to the frame. When the gage is connected to the gage point, fluid under pressure enters the tube. On an increase in internal pressure, the elliptical cross-section of the tube is transmitted, through a link quadrant and pinion to a pointer which by moving clockwise on this graduated circular dial indicator the pressure intensity of the fluid. The dial of the gage is so calibrated that it reads zero when the pressure inside the tube equals the local atmospheric pressure, and the elastic deformation of the tube causes the pointer to be displaced on the dial in proportion to the pressure intensity of the fluid. By using the tube of appropriate stiffness, gages for wide range of pressure may be made.

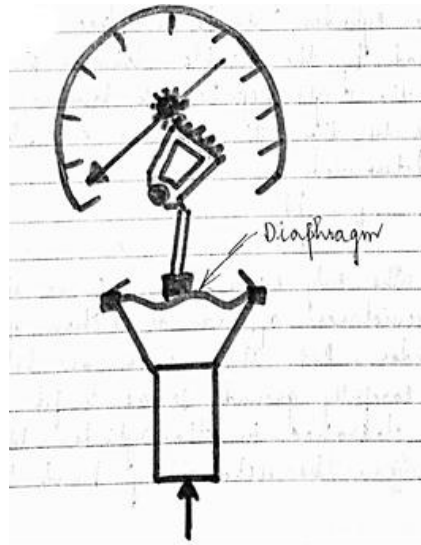


Further, by suitably modifying the graduations of the dial and adjusting the pointer, Bourdon tube vacuum gages can also be made. When a vacuum gage is connected to a partial vacuum, the tube tends to close, thereby moving the pointer in anti-clockwise direction, indicating the negative or vacuum pressure. The gage dials are usually calibrated to read newton per square metre (N/m^2) or pascal (Pa), or kilogram (f) per square centimetre [$\text{kg(f)}/\text{cm}^2$]. However, other units of pressure, such as metres of water or centimetres of mercury, are also frequently used.

(ii) Diaphragm Pressure Gage

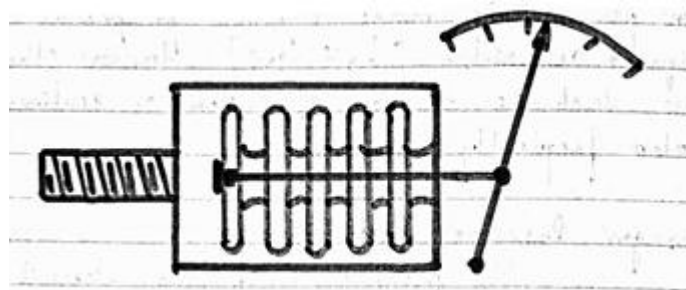
The pressure responsive element in this gage is an elastic steel corrugated diaphragm. The elastic deflection of the diaphragm under pressure is transmitted to a pointer by a suitable arrangement as in the case of Bourdon tube pressure gage. However, this gage is used to measure

relatively low pressure intensities. The aneroid barometer operates on a similar principle.



(iii) Bellows Pressure Gage

In this gage, the pressure responsive element is made up of a thin metallic tube having deep circumferential corrugations. In response to the pressure changes, this elastic element expands or contracts, thereby moving the pointer on a graduated circular dial.

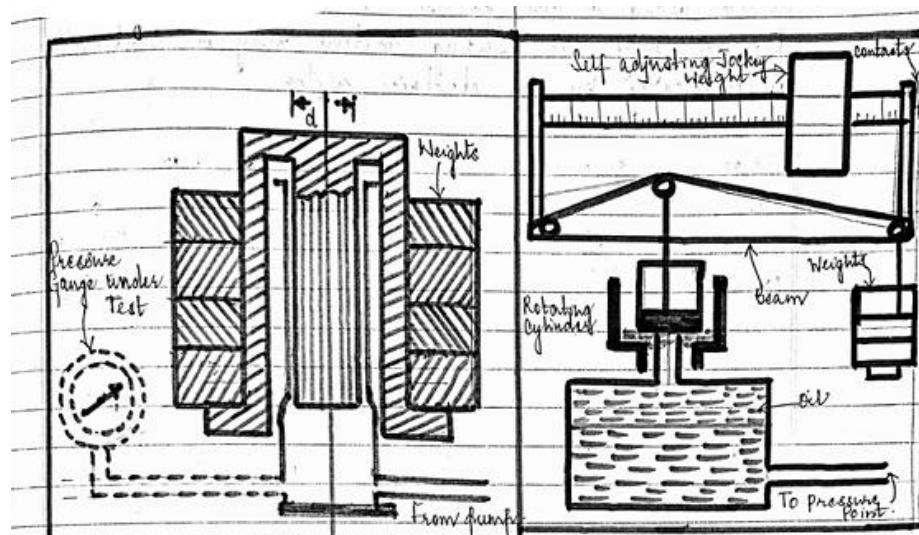


(iv) Dead-Weight Pressure Gage

A simple form of a dead-weight pressure gage consists of a plunger of diameter d , which can slide within a vertical cylinder. The fluid under pressure, entering the cylinder, exerts a force on the plunger, which is balanced by the weights loaded on the top of the plunger. If the weight required to balance the fluid under pressure is W , then the pressure intensity of the fluid may be determined as:

$$p = \frac{W}{\frac{\pi}{4} d^2}$$

The only error that may be involved is due to frictional resistance offered to motion of the plunger in the cylinder. But this error can be avoided if the plunger is carefully ground, so as to fit with the least permissible clearance in the cylinder. Moreover, the whole mass can be rotated by hand before final readings are taken



Dead-weight gages are generally not used so much to measure the pressure intensity at a particular point as to serve as standards of comparison. Hence, as shown in figure, a pressure gage which is to be checked or calibrated is set parallel with the dead-weight gage. Oil under pressure is pumped into the gages, thereby lifting the plunger and balancing it against the oil pressure by loading with known weights. The pressure intensity of the oil being thus known, the attached pressure gage can either be tested for its accuracy or it can be calibrated.

A dead-weight gage which can be used for measuring pressure at a point with more convenience is also shown in figure. In this gage a lever, some or all of the weighing machines, is provided to magnify the thrust of the pressure. The load required to balance the force due to fluid pressure is first roughly adjusted by hanging weights from the end of the main beam. Then a smaller jockey weight is slid along to give precise balance. In more precise type of gage the sliding motion may be contrived automatically by an electric motor.